

Polymer Plant Scheduling — PE & PP Production Campaigns

Manufacturing scheduling case study · Julia + JuMP + HiGHS · companion to `notebooks/polymer_pe_pp_scheduling.jl`

This report documents a **production scheduling model** for a single compounding / extrusion line that manufactures several grades of **polyethylene (PE)** and **polypropylene (PP)**. Switching the line between grades — and especially between the PE and PP **families** — costs time and money: purging, catalyst/additive changes, and off-spec transition material. The model below decides the order in which to run a set of confirmed orders so as to minimize total changeover cost plus tardiness cost. It is a mixed-integer linear program (MILP), built with [JuMP.jl](#) and solved with the open-source [HiGHS](#) solver.

1. Plant data: grades, families, and orders

The line currently has **7 confirmed orders** across **5 grades**: three PE grades (HDPE, LLDPE, LDPE) and two PP grades (homopolymer and copolymer). Each order has a quantity, a production rate on this line, and a customer due date (in hours from the scheduling horizon start, $t = 0$).

Order	Grade	Family	Qty (t)	Rate (t/h)	Due (h)	Weight
1	HDPE-5502	PE	120	5.0	30	1.0
2	LLDPE-2020	PE	80	4.5	40	1.0
3	PP-Homo-1100	PP	100	5.5	55	1.5
4	LDPE-1922	PE	60	4.0	65	1.0
5	PP-Copo-3300	PP	90	4.8	80	1.5
6	HDPE-5502	PE	70	5.0	95	1.0
7	PP-Homo-1100	PP	110	5.5	110	1.5

2. Sequence-dependent changeovers

Because PE and PP require different catalysts and purge procedures, the changeover time depends on *what* the line just ran, not just on the next grade:

Transition	Time (h)	Why
Same grade repeated	0.5	housekeeping only
Same family, different grade	2.5	color/additive change, partial purge
PE ↔ PP family switch	7.0	full line/reactor purge + catalyst change

Cost rates used in this run: changeover **\$450/h**, tardiness **\$300/h** per unit order weight. (The companion Pluto notebook exposes both as live sliders.)

3. MILP formulation

Node 0 is a dummy "line idle" start state; nodes 1..n are the orders. Binary variables x_{ij} indicate that order j runs immediately after node i . Continuous variables C_i and T_i are the completion time and tardiness of order i .

```

minimize    c_co * sum_{i != j} s_ij * x_ij  +  c_tar * sum_i w_i * T_i

subject to  sum_j x_0j = 1
            sum_{i != j} x_ij = 1                for all j
            sum_{j != i} x_ij ≤ 1                for all i
            C_j ≥ C_i + s_ij + p_j - M(1 - x_ij)    for all i != j
            T_i ≥ C_i - d_i,    T_i ≥ 0
            x_ij ∈ {0,1},    C_i, T_i ≥ 0

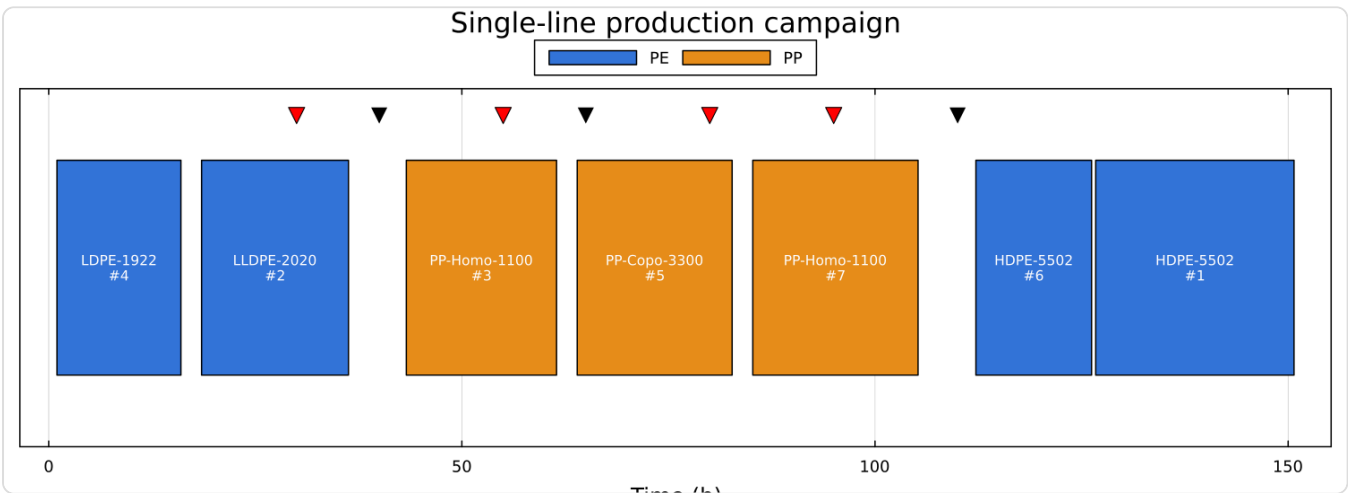
```

Completion times must strictly increase along any chain of active arcs, so this "big-M timing" formulation rules out sub-tours on its own — no separate subtour-elimination constraints (e.g. MTZ) are needed for a single open path. With 7 orders the model has 49 binary variables and solves to global optimality with HiGHS in well under a second.

4. Optimal campaign sequence

Order	Grade	Family	Qty (t)	Start (h)	End (h)	Due (h)	Status
4	LDPE-1922	PE	60	1.0	16.0	65.0	on time
2	LLDPE-2020	PE	80	18.5	36.3	40.0	on time
3	PP-Homo-1100	PP	100	43.3	61.5	55.0	6.5 h late
5	PP-Copo-3300	PP	90	64.0	82.7	80.0	2.7 h late
7	PP-Homo-1100	PP	110	85.2	105.2	110.0	on time
6	HDPE-5502	PE	70	112.2	126.2	95.0	31.2 h late
1	HDPE-5502	PE	120	126.7	150.7	30.0	120.7 h late

5. Gantt chart



Grade blocks are colored by polymer family (**PE** vs. **PP**); triangles mark each order's due date (red = missed). The optimizer clusters same-family grades together to avoid the expensive 7-hour PE ↔ PP purge, trading a little lateness on lower-weight orders when that is cheaper than an extra family switch.

6. Key performance indicators

MAKESPAN	CHANGEOVER TIME	TARDINESS	TOTAL COST
150.7 h	23.0 h	161.1 h (4 ord.)	\$60052

7. Discussion & extensions

This single-line, deterministic model already captures the dominant economics of PE/PP campaign scheduling — family-driven changeovers and due-date pressure — while staying small enough to solve to global optimality in well under a second. Natural next steps for a production-grade version:

- **Multiple parallel lines** — extend x to a 3-index $x[i,j,line]$ and add line-capability restrictions (not every line can run every grade).
- **Minimum/maximum campaign length** — avoid uneconomically short runs by bounding the number of consecutive orders of the same grade.
- **Exact grade-pair changeover matrix** — replace the family-level matrix with real historical purge times per ordered grade pair.
- **Inventory & storage costs** — silo capacity limits often force a grade off the natural due-date order even when changeovers are cheap.
- **Rolling-horizon re-optimization** — re-solve as new orders arrive or priorities change, warm-starting from the current sequence.
- **Robustness** — sample uncertain processing rates / late raw-material arrivals and re-optimize, or solve a chance-constrained variant.

Because the model is expressed as a compact JuMP program, all of the above are incremental changes to the constraints already built above, not a rewrite. An interactive version of this same model — with live cost-weight sliders — is available as a Pluto.jl notebook at `notebooks/polymer_pe_pp_scheduling.jl` in this repository.